

DANA 4840

Classification II

Cluster Analysis

Part 3

Distance (dissimilarity) measures

When we deal with quantitative data, the dissimilarity of 2 objects is quantified by the distance between their 2 vectors of variables (of length n):

- **Euclidean distance formula:** $\sqrt{\sum (x_i - y_i)^2}$
- **Manhattan distance formula:** $\sum |(x_i - y_i)|$

Distance (dissimilarity) measures

Example:

	DANA 4800	DANA 4820	DANA 4840
Student A	68	71	70
Student B	80	81	90

- (a) Calculate the Euclidean distance between student A and student B.
- (b) Calculate the Manhattan distance between student A and student B.

Distance (dissimilarity) measures

- Naturally, a change in the units (of the variables) will change the relative importance of the variables. Hence, algorithms (such as k-means) is often applied to variables that have been scaled. In R, use the `scale()` function.
- In literature, we learnt that Manhattan distance is less affected by outliers than Euclidean distance.

Correlation-based distances

- If you wish to identify clusters of observations that go “up” or “down” together regardless of their magnitudes, then you should consider using **correlation-based distances** as a dissimilarity measure.
- An example when correlation-based distance might be relevant:

Which pair of students are more alike in their spending habits?

	Food	Rent	Clothes	Toiletries
Student A	\$500	\$1200	\$500	\$200
Student B	\$700	\$350	\$700	\$180
Student C	\$20	\$48	\$20	\$8

Correlation-based distances

- If x , y can be assumed to have a linear relationship, we can use **Pearson's correlation distance**.
- Pearson's correlation distance is the most commonly-used correlation-based distance.
- However, Pearson's correlation is sensitive to outliers. If that is a concern, you can consider using **rank-based correlations** such as Spearman correlation distance and Kendall correlation distance.

Some useful R commands

- Calculate Euclidean Distance using R:

Use `dist()`

- Calculate Correlation-based distance:

Use `get_dist()` in factoextra package

- Calculate distances for mixed data:

Use `daisy()` in cluster package

- Visualize distance matrices:

Use `fviz-dist()` in factoextra package